

Q1. JEE Main 2026 (21 January Shift 2)

Let the maximum value of $(\sin^{-1} x)^2 + (\cos^{-1} x)^2$ for $x \in \left[-\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}\right]$ be $\frac{m}{n}\pi^2$, where $\gcd(m, n) = 1$. Then $m + n$ is equal to ____.

Q2. JEE Main 2026 (28 January Shift 2)

Considering the principal values of inverse trigonometric functions, the value of the expression

$\tan\left(2\sin^{-1}\left(\frac{2}{\sqrt{13}}\right) - 2\cos^{-1}\left(\frac{3}{\sqrt{10}}\right)\right)$ is equal to :

- (1) $\frac{16}{63}$ (2) $-\frac{33}{56}$ (3) $-\frac{16}{63}$ (4) $\frac{33}{56}$

Q3. JEE Main 2026 (24 January Shift 2)

If the domain of the function $f(x) = \sin^{-1}\left(\frac{1}{x^2-2x-2}\right)$, is $(-\infty, \alpha] \cup [\beta, \gamma] \cup [\delta, \infty)$, then $\alpha + \beta + \gamma + \delta$ is equal to

- (1) 2 (2) 3 (3) 4 (4) 5

Q4. JEE Main 2026 (22 January Shift 1)

The number of solutions of $\tan^{-1} 4x + \tan^{-1} 6x = \frac{\pi}{6}$, where $-\frac{1}{2\sqrt{6}} < x < \frac{1}{2\sqrt{6}}$, is equal to

- (1) 3 (2) 0 (3) 2 (4) 1

Q5. JEE Main 2026 (21 January Shift 1)

If the domain of the function $f(x) = \cos^{-1}\left(\frac{2x-5}{11-3x}\right) + \sin^{-1}(2x^2 - 3x + 1)$ is the interval $[\alpha, \beta]$, then $\alpha + 2\beta$ is equal to :

- (1) 3 (2) 5 (3) 1 (4) 2

Q6. JEE Main 2026 (28 January Shift 1)

If $k = \tan\left(\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\left(\frac{2}{3}\right)\right) + \tan\left(\frac{1}{2}\sin^{-1}\left(\frac{2}{3}\right)\right)$, then the number of solutions of the equation

$\sin^{-1}(kx - 1) = \sin^{-1} x - \cos^{-1} x$ is ____

ANSWERS AND SOLUTIONS

1. 65

2. (4)

3. (3)

4. (4)

5. (1)

6. 1

1. (65) Let $\alpha = \sin^{-1} x$ and $\beta = \cos^{-1} x$. Using $\alpha + \beta = \frac{\pi}{2}$:

$$f(x) = \alpha^2 + \beta^2 = \alpha^2 + \left(\frac{\pi}{2} - \alpha\right)^2 = 2\alpha^2 - \pi\alpha + \frac{\pi^2}{4}$$

This equals $2\left(\alpha - \frac{\pi}{4}\right)^2 + \frac{\pi^2}{8}$, which is minimized at $\alpha = \frac{\pi}{4}$.

For $x \in \left[-\frac{\sqrt{3}}{2}, \frac{1}{\sqrt{2}}\right]$, we have $\alpha \in \left[-\frac{\pi}{3}, \frac{\pi}{4}\right]$.

Distance from $-\frac{\pi}{3}$ to vertex: $\frac{\pi}{4} + \frac{\pi}{3} = \frac{7\pi}{12}$. Distance from $\frac{\pi}{4}$ to vertex: 0.

Maximum occurs at $x = -\frac{\sqrt{3}}{2}$ where $\alpha = -\frac{\pi}{3}$:

$$f\left(-\frac{\sqrt{3}}{2}\right) = 2\left(\frac{\pi^2}{9}\right) + \frac{\pi^2}{3} + \frac{\pi^2}{4} = \pi^2\left(\frac{8+12+9}{36}\right) = \frac{29}{36}\pi^2$$

Since $\gcd(29, 36) = 1$, we have $m = 29, n = 36$, so $m + n = 65$

2. (4) Let $\alpha = \sin^{-1}\left(\frac{2}{\sqrt{13}}\right)$.

Then $\sin \alpha = \frac{2}{\sqrt{13}}, \cos \alpha = \frac{3}{\sqrt{13}}, \tan \alpha = \frac{2}{3}$. So $\tan(2\alpha) = \frac{2 \cdot \frac{2}{3}}{1 - \frac{4}{9}} = \frac{12}{5}$.

Let $\gamma = \cos^{-1}\left(\frac{3}{\sqrt{10}}\right)$.

Then $\cos \gamma = \frac{3}{\sqrt{10}}, \sin \gamma = \frac{1}{\sqrt{10}}, \tan \gamma = \frac{1}{3}$. So $\tan(2\gamma) = \frac{2 \cdot \frac{1}{3}}{1 - \frac{1}{9}} = \frac{3}{4}$.

$$\text{Therefore } \tan(2\alpha - 2\gamma) = \frac{\frac{12}{5} - \frac{3}{4}}{1 + \frac{12}{5} \cdot \frac{3}{4}} = \frac{\frac{33}{20}}{\frac{56}{20}} = \frac{33}{56}$$

3. (3) For $\sin^{-1}\left(\frac{1}{x^2-2x-2}\right)$ to be defined, we need $\left|\frac{1}{x^2-2x-2}\right| \leq 1$, i.e., $|x^2 - 2x - 2| \geq 1$.

Case 1:

$$x^2 - 2x - 2 \geq 1 \Rightarrow x^2 - 2x - 3 \geq 0$$

$$\Rightarrow (x-3)(x+1) \geq 0$$

$$\Rightarrow x \leq -1 \text{ or } x \geq 3.$$

Case 2:

$$x^2 - 2x - 2 \leq -1 \Rightarrow x^2 - 2x - 1 \leq 0$$

$$\Rightarrow (x-1)^2 \leq 2$$

$$\Rightarrow 1 - \sqrt{2} \leq x \leq 1 + \sqrt{2}.$$

$$\text{Domain} = (-\infty, -1] \cup [1 - \sqrt{2}, 1 + \sqrt{2}] \cup [3, \infty).$$

$$\text{So } \alpha = -1, \beta = 1 - \sqrt{2}, \gamma = 1 + \sqrt{2}, \delta = 3.$$

$$\alpha + \beta + \gamma + \delta = -1 + 1 - \sqrt{2} + 1 + \sqrt{2} + 3 = 4.$$

4. (4) For $x \in \left(-\frac{1}{2\sqrt{6}}, \frac{1}{2\sqrt{6}}\right)$, $24x^2 < 1$

$$\text{So } \tan^{-1}(4x) + \tan^{-1}(6x) = \tan^{-1}\left(\frac{10x}{1-24x^2}\right) = \frac{\pi}{6}.$$

$$\frac{10x}{1-24x^2} = \frac{1}{\sqrt{3}} \Rightarrow 24x^2 + 10\sqrt{3}x - 1 = 0.$$

$$x = \frac{-10\sqrt{3} \pm \sqrt{300+96}}{48} = \frac{-5\sqrt{3} \pm 3\sqrt{11}}{24}.$$

$$x_1 = \frac{-5\sqrt{3} + 3\sqrt{11}}{24} \approx 0.054 \text{ (valid, lies in range).}$$

$$x_2 = \frac{-5\sqrt{3} - 3\sqrt{11}}{24} \approx -0.776 \text{ (rejected, outside range).}$$

Number of solutions = 1.

5. (1) For $\cos^{-1}\left(\frac{2x-5}{11-3x}\right)$: $-1 \leq \frac{2x-5}{11-3x} \leq 1$ gives $x \leq \frac{16}{5}$ or $x > 6$.

For $\sin^{-1}(2x^2 - 3x + 1)$: $-1 \leq 2x^2 - 3x + 1 \leq 1$.

Lower bound always satisfied (discriminant < 0).

Upper bound gives $x(2x - 3) \leq 0$, so $0 \leq x \leq \frac{3}{2}$.

Intersection of both conditions: $[0, \frac{3}{2}]$.

Thus $\alpha = 0$, $\beta = \frac{3}{2}$.

$\alpha + 2\beta = 0 + 3 = 3$.

6. (1) Let $\theta = \frac{1}{2}\sin^{-1}\frac{2}{3}$, then $\frac{1}{2}\cos^{-1}\frac{1}{3} = \left(\frac{\pi}{4} - \theta\right)$

$$k = \tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta} = \frac{1}{\sin 2\theta}$$

$$k = \frac{2}{\frac{3}{2}} = 3$$

$$\sin^{-1}(3x - 1) = \sin^{-1}x - \cos^{-1}x$$

$$\sin^{-1}(3x - 1) = \frac{\pi}{2} - 2\cos^{-1}x$$

$$3x - 1 = \sin\left(\frac{\pi}{2} - 2\cos^{-1}x\right)$$

$$3x - 1 = 2x^2 - 1 \Rightarrow x = 0, \frac{3}{2} \text{ (rejected)}$$

No. of solution = 1